

# CAD FIT CHECK - will the real parts fit the printed parts?

**Date:** 2026-07-29 **Scope:** physical fit of purchased/owned components inside the v7.2 printed parts. **Status of the build:** nothing has ever been assembled. This is the last check before spending.

## How to read this document

Every statement is tagged:

- **[SCAD]** - read directly out of the OpenSCAD source in `cad/scad/`. This is fact about the \*printed part\*.
- **[SPEC]** - a published nominal figure for a standard catalogue part (28BYJ-48, 6x6 tactile switch, 5.5/2.1 barrel jack, micro-USB). Standard parts still vary between clone vendors.
- **[ASSUMED]** - a number somebody typed into the CAD about a physical object that has never been measured. Treat as fiction until a caliper says otherwise.

**Nothing in the "Real part needs" column below is a measured fact about Mridul's actual parts.** Only the 8x1mm docking magnets have ever been physically measured (`docs/SOURCING.md`, v6.1).

## 1. SUMMARY TABLE

| Component                                      | CAD provides [SCAD]                                                                                                                          | Real part needs                                                     | Verdict                      | Action                                                                                   |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------|
| Hall sensor (MH-Sensor-Series / KY-024 module) | 5.3 x 4.3 x 3.0 mm pocket at (0, +20) in <code>base_plate</code>                                                                             | PCB module ~15 x 11 x 1.6 mm + ~11 mm of 4-pin header [ASSUMED]     | <b>FAILS</b>                 | Desolder the bare sensor, or move the module out of the plate entirely                   |
| Hall pocket itself                             | pocket floor <code>z=2</code> ; dia 46 cam pocket floor also <code>z=2</code> , and the pocket lies <b>entirely inside</b> <code>r=23</code> | any pocket at all                                                   | <b>FAILS</b>                 | The pocket is 100% subsumed by the cam pocket - it does not exist as a feature           |
| Cam homing magnet dia 3x2                      | dia 3.2 x 2.0 pocket at <code>r=17.35</code> in a <b>2.0 mm</b> disc floor                                                                   | 3x2 mm disc magnet - <b>not owned</b>                               | <b>FAILS</b>                 | Pocket is a through-hole that punches a dia 3.2 crater through cam <b>tracks 2 and 3</b> |
| 28BYJ-48 motor body                            | dia 29 seat, 35 mm hole spacing, dia 4.3 holes, body centre at <code>x=-8</code> , 19 mm of <code>z-space</code>                             | body dia 28, 35 mm spacing, dia 4.2, ~8 mm offset, 19 mm can [SPEC] | <b>UNVERIFIED</b>            | Never measured. 8 caliper readings required (\$4)                                        |
| Motor right mounting screw                     | hole at (+9.5, 0) falls <b>inside</b> the 22 x 16 spring-cavity through-window                                                               | material to clamp against                                           | <b>FAILS</b>                 | Motor can only be held by ONE screw                                                      |
| Motor screw head vs cam                        | left hole at <code>x=-25.5</code> is clear; right hole at <code>x=+9.5</code> is under the dia 46 cam disc                                   | screw head clearance                                                | <b>FAILS</b>                 | No counterbore modelled; head fouls the cam floor                                        |
| Cam on shaft (double-D bore)                   | dia 5.2 clipped to 3.2 flats; <b>through-hole</b> , real engagement = hub 4 + floor 2 = <b>6 mm</b>                                          | 5.0 mm shaft, 3.0 mm across flats, ~10 mm long [SPEC]               | <b>UNVERIFIED / marginal</b> | Clearances fine on paper; <code>z-stack</code> is wrong (\$5)                            |
| Cam z-stack vs motor face                      | hub bottom <code>z=39</code> , motor top face <code>z=41</code>                                                                              | hub cannot go below the motor face                                  | <b>FAILS</b>                 | Cam bottoms on the motor and sits ~2 mm high -> all six dots permanently proud           |
| Shaft protrusion                               | cam surface <code>z=45</code> ; a 10 mm shaft reaches <code>z=51</code>                                                                      | shaft must not enter the dot cluster                                | <b>FAILS (conditional)</b>   | Shaft passes through the open bore into the linkage risers at <code>x=+/-2.4</code>      |
| ULN2003 module, wires soldered flat            | 36 x 46 x 16 pocket + 2 mm mid-plate relief slot = 18 mm                                                                                     | ~35 x 32 x ~12 mm [ASSUMED, established earlier]                    | <b>FITS</b>                  | Solder wires flat. Watch the 0.5 mm/side X clearance                                     |
| ULN2003 module, vertical Dupont jumpers        | same 18 mm max                                                                                                                               | ~20 mm as-wired [ASSUMED]                                           | <b>FAILS</b>                 | Confirmed: 20 > 18. No vertical jumpers                                                  |

| Component                             | CAD provides [SCAD]                                                                                             | Real part needs                                                                     | Verdict                      | Action                                                                                               |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------|
| <b>ULN2003 vs muscle-board bosses</b> | 4 mm tall bosses at (+/-14, +/-19) rise from the pocket floor                                                   | flat floor                                                                          | <b>UNVERIFIED</b>            | If the board is ~35 mm in Y it clips the y=+/-19 bosses                                              |
| <b>ESP32 DevKit length in pod</b>     | cavity +/-28 in X; board 51.5 centred at x=+4 -> spans -21.75 .. <b>+29.75</b>                                  | board must end inside x=+28                                                         | <b>FAILS</b>                 | 1.75 mm interference with the +X (dock) wall                                                         |
| <b>ESP32 DevKit width / row pitch</b> | devkit_width 28.0, hdr_row_pitch 25.4 [ASSUMED]                                                                 | 30-pin DOIT V1 is commonly ~25.4 wide / 0.9" pitch; 38-pin is ~28 / 1.0" [SPEC-ish] | <b>UNVERIFIED</b>            | The CAD numbers describe the <b>38-pin</b> board; the docs say 30-pin                                |
| <b>USB cutout alignment</b>           | cutout z 12.5 .. 19.5; board top sits at z=10.5 (8.5 strip ? 1.0 channel recess)                                | port body ~10.5 .. 13.0                                                             | <b>FAILS</b>                 | Hole sits ~2 mm too high; only ~0.5 mm overlaps the port                                             |
| <b>USB cutout size</b>                | 10 x 7 mm                                                                                                       | micro-USB overmould commonly ~12 x 8 [SPEC-ish]                                     | <b>UNVERIFIED</b>            | Plug may not enter                                                                                   |
| <b>Headroom under pod lid</b>         | board top z~10.5, cavity top z=54                                                                               | a few mm                                                                            | <b>FITS</b>                  | 43 mm spare - grossly over-tall, but no clash                                                        |
| <b>Barrel jack hole</b>               | dia 11.5 at (-22, +18) on the lid                                                                               | 5.5/2.1 panel jack, ~dia 11 thread [SPEC]                                           | <b>UNVERIFIED</b>            | User's jack is a yellow screw-terminal type, not the modelled panel jack                             |
| <b>Barrel jack cradle</b>             | inner 12 x 9.5 x 11, walls 1.5 -> outer reaches x = <b>-29.5</b>                                                | must fit inside cavity x >= -28                                                     | <b>FAILS</b>                 | 1.5 mm interference - the lid will not close. And all three dims are PLACEHOLDERS                    |
| <b>Docking magnets</b>                | dia 8.4 x 1.2 pockets, 2 per face at y=+/-14, 4 mm wall                                                         | 8 x 1 mm NdFeB discs - <b>measured, in hand</b>                                     | <b>FITS</b>                  | Only re-confirm the newly bought batch is the same 8x1                                               |
| <b>Nav buttons - cap stem</b>         | dia 3.8 stem projects <b>4.5 mm behind</b> the flange through a dia 4.2 hole                                    | must cross a 4 mm wall                                                              | <b>FITS (nominal)</b>        | Stem reaches 0.5 mm inside; 0.4 mm diametral clearance allows resin-in-PETG sliding                  |
| <b>Nav buttons - switch reach</b>     | The 6 mm switch body is stopped by the pod's inner front wall; cage back wall takes press load                  | plunger must meet the stem                                                          | <b>FITS (nominal)</b>        | The front wall is the forward datum; bench-test the actual switch's plunger before final resin batch |
| <b>6x6x5 tactile switch in cage</b>   | 6.4 mm slot, 7.2 mm depth, legs vertical                                                                        | 6.0 x 6.0 x 5.0 body [SPEC]                                                         | <b>FITS</b>                  | Body and revised cap reach fit on the nominal switch geometry                                        |
| <b>Dot insert into top plate</b>      | body 11.0 into 11.2 opening; flange 15.0 into 15.2 rebate; heights 2.0 / 1.2; glue shelf 2.0 mm wide, 105.6 mm2 | resin tile printed from the same file                                               | <b>FITS</b>                  | Verified numerically. Only risk is FDM square holes printing undersize                               |
| <b>Base-plate standoffs</b>           | dia 6 posts at x=+/-26, plate half-length 28 -> 1 mm hangs off the edge                                         | supported footprint                                                                 | <b>UNVERIFIED / cosmetic</b> | Prints as a small unsupported overhang; not structural                                               |

## Required outer-box changes - handout checklist

These items are deliberately **not** applied to `outer_box.scad` yet. They either require the real motor dimensions or a hardware-choice decision; changing them from catalogue numbers would simply move the fit failure somewhere else.

| Priority | Required change                                                    | Why                                                                                                          | Prerequisite                                          |
|----------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Blocker  | Re-derive <code>base_plate_z</code> and the motor-height allowance | The current 41 mm motor-face height is an unmeasured assumption. It controls the cam/linkage vertical stack. | Measure the actual motor can height and shaft height. |

| Priority                                         | Required change                                                                                              | Why                                                                                                    | Prerequisite                                                                                                 |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Blocker                                          | Redesign the base-plate spring cavity around both motor ears                                                 | The right motor screw opens into the existing 22 x 16 mm through-window, leaving a one-screw mount.    | Measure the actual motor mounting-ear envelope and hole centres.                                             |
| Blocker                                          | Widen/reposition the base-plate support footprint if the motor-ear fix needs it                              | The left ear currently has only a 0.35 mm edge wall and the base standoffs overhang the plate by 1 mm. | Motor-ear measurement, then confirm the revised plate still clears the 60 mm shell cavity and corner bosses. |
| Required before installing the owned Hall module | Add a real Hall-module bracket in the outer-box electronics area, or commit to a measured bare-sensor design | The current base-plate pocket is entirely subsumed by the cam pocket and cannot hold the owned module. | Decide module versus desoldered sensor; measure its footprint and sensing-face offset.                       |
| Decision required                                | Remove/relocate the four future-muscle-board bosses when using the ULN2003 module                            | A 35 mm-wide ULN2003 can collide with the y= $\pm$ 19 mm bosses in the outer-box electronics pocket.   | Measure the actual ULN2003 board and choose prototype driver versus future muscle PCB.                       |

**Do not alter the 68 x 68 mm shell, docking windows, magnet pockets, or 4 mm walls as part of this repair.**

Those features are independent of the motor and have already been physically fit-tested. The next safe sequence is: measure hardware -> update the dependent base/mid/cam/ outer-box dimensions together -> render the full assembly -> print a cheap proving part.

## 2. HALL SENSOR - FAILS, twice over

### 2a. The pocket does not physically exist

[SCAD] base\_plate.scad:

- hall\_pocket\_h = 3, cut from base\_thickness - hall\_pocket\_h = **z = 2** upward.
- cam\_pocket\_depth = 3, cam\_pocket\_diameter = 46, cut from **z = 2** upward, radius 23.
- Hall pocket footprint: x ? [-2.65, +2.65], y ? [17.85, 22.15]. Its farthest corner is at

radius ?(2.652 + 22.152) = **22.31 mm < 23 mm**.

The hall pocket lies **entirely inside the cam pocket, at exactly the same depth**. Subtracting it removes nothing. There is no rectangular recess on the printed part - just flat cam-pocket floor.

Worse: the cam disc's underside \*sits on that floor\* (cam\_flat\_z = 45, disc floor 2 mm, pocket floor z=43 in box coordinates). There is **zero vertical space** above the "pocket" for any sensor, bare TO-92 or otherwise. The wire channel (hall\_wire\_w=2 x hall\_wire\_d=1) does survive, but only in the strip of plate outside r=23.

### 2b. The module is far too big anyway

[ASSUMED - MEASURE THIS] The blue MH-Sensor-Series / KY-024-style board Mridul owns (identified by Codex on 2026-06-03 as an analog+digital module with AO/DO/GND/VCC) is a PCB in the region of **15 x 11 mm, ~1.6 mm thick, plus a 4-pin 2.54 mm header adding roughly 8-11 mm of height**. Those figures are typical for that family of modules; they are **not** measured from his part.

Against a 5.3 x 4.3 x 3.0 pocket sized for a bare SS49E/A3144 TO-92 body:

- length: needs ~15, has 5.3 -> **~10 mm short**
- width: needs ~11, has 4.3 -> **~7 mm short**
- height: needs ~10-13 with the header, has 3 (and really has 0, see 2a) -> **hopeless**

### Fixes, in order of preference

**1 Desolder the bare sensor from the module** (recommended). The MH-series board carries a standard TO-92 hall device plus an LM393 comparator, an LED and a trimpot. Unsolder the three legs, mount the bare TO-92 in the plate, and run three wires back to the module (or straight to the ESP32 with a pull-up, since the breadboard test already ran the analog output at 3.3 V). This keeps the CAD footprint honest - but §2a still has to be fixed first, because the pocket currently isn't there.

**1 Move the module off the base plate entirely.** Mount it on the electronics-pocket floor or on a wall of the outer box, with the sensing face pointing up at the cam, and run wires. The cam magnet is 3 mm dia - flux falls off fast, so this only works if the module can get within a few mm of the magnet path. This is the lowest-CAD-risk

option but needs a new bracket.

**1 Enlarge and deepen the pocket.** Requires all three of: (a) make `hall_pocket_h > 3` so the pocket floor drops below the cam pocket floor (e.g. 4.0-4.5 mm, leaving 0.5-1.0 mm of plate underneath); (b) grow `hall_pocket_w/d` to the measured module footprint + 0.4 mm; (c) move `hall_pocket_y` outboard past `y=23` so it is no longer under the disc, **or** add a header escape slot. Note the plate is only 5 mm thick and the +Y edge is at `y=25` - there is very little room. Option 1 or 2 is more realistic.

### 2c. BONUS DEFECT FOUND - the cam homing-magnet pocket is a hole through two tracks

**[SCAD]** `braille_cam.scad`: `magnet_dia=3.0`, `magnet_depth=2.0`, cut as `cylinder(d=3.2, h=3)` from `z=-1`, i.e. `z = -1 .. +2`, at `r = 17.35`, angle 90deg. The disc floor is `disk_base_thickness = 2.0`, spanning `z = 0 .. 2`. The cut therefore removes the **entire floor thickness** - it is a through-hole, not a pocket ("leaves 0mm floor" is even in the comment). And at `r = 17.35 +/- 1.6` it spans `r = 15.75 .. 18.95`, which overlaps **track 2** (`r 15.4-17.0`, drives dot 1) and **track 3** (`r 17.1-18.7`, drives dot 2).

Where those tracks are in their DOWN state the cam surface at that angle is simply **missing** - a dia 3.2 mm hole in a running surface that a 1.0 mm-thick linkage foot crosses once per revolution. The foot will drop in and jam.

**Fix:** shrink `magnet_depth` to ~1.4 mm and thicken the local floor, or (better) move `magnet_radius` inboard to `r < 12` (inside `inner_radius`, under no track at all) and move the hall sensor to match. Moving the magnet inboard also weakens coupling, so re-check the sensor gap. Also: **the 3x2 mm homing magnet is not in hand** (`docs/SOURCING.md §2`) - it still has to be bought.

## 3. MOTOR - NEVER MEASURED. Treat every motor number as fiction.

[!] **THE 28BYJ-48 IN THIS PROJECT HAS NEVER BEEN TOUCHED WITH A CALIPER.** `base_plate.scad`, `mid_plate.scad` and `braille_cam.scad` all encode motor dimensions that were typed from datasheets and forum posts. Mridul has photographed a motor whose shaft looked different from the model. Every verdict in this section is conditional on measurement.

**[SCAD] what the CAD assumes:**

| Parameter                     | Value                                                          | File                                                       |
|-------------------------------|----------------------------------------------------------------|------------------------------------------------------------|
| body diameter                 | dia 29 (seat pocket, 1.5 mm deep)                              | <code>base_plate.scad</code>                               |
| retaining collar ID           | dia 29.5, 8 mm tall, at <code>x=-8</code>                      | <code>mid_plate.scad</code>                                |
| mount-hole spacing            | 35 mm                                                          | <code>base_plate.scad</code>                               |
| mount-hole diameter           | dia 4.3                                                        | <code>base_plate.scad</code>                               |
| shaft offset from body centre | -8 mm (shaft at <code>x=0</code> , body at <code>x=-8</code> ) | <code>base_plate.scad</code> , <code>mid_plate.scad</code> |
| can height allowance          | 19 mm ( <code>z=22 .. 41</code> )                              | <code>outer_box.scad</code> <code>base_plate_z</code>      |
| shaft                         | dia 5.0, 3.0 across flats, 10 mm long                          | <code>braille_cam.scad</code> comments                     |

**[SPEC]** The published nominal 28BYJ-48 is: can dia 28 x 19 mm, mounting bracket holes 35 mm apart at dia 4.2, shaft offset ~8 mm from the can centre, shaft dia 5.0 with two flats ~3.0 mm apart, protruding ~10 mm. So the CAD is \*plausible\*. Clone variance and the 19 mm z-budget with **zero** slack are the risks.

### Two hard failures that exist regardless of measurement

**3a. The right-hand motor screw has nothing to screw into.** **[SCAD]** `spring_cavity()` is `cube([22, 16, 20], center=true)` - a through-window spanning `x ? [-11, +11]`, `y ? [-8, +8]`. The right mounting hole is at `x = -8 + 35/2 = +9.5`, `y = 0`, dia 4.3 -> spans `x ? [7.35, 11.65]`. That is inside the window. There is no plate material there. **The motor can only be bolted by its left ear (`x=-25.5`).** A single-screw stepper will rock and lose steps.

**3b. The left ear hole is 0.35 mm from the plate edge.** `x = -25.5`, dia 4.3 -> outer edge at -27.65; plate half-length is 28. A 0.35 mm wall on a 0.4 mm nozzle is not a wall. The screw will break out.

**Fix for both:** shrink/reshape `spring_cavity` so it clears `x=+9.5` (e.g. two smaller windows, or move it to -Y), and widen `base_length` from 56 to at least 60 (the box cavity is 60 - so this also needs the cavity or the standoff pattern revisited).

### Measurements required (feeds -> CAD parameter)

**1** Can outer diameter -> `motor_body_diameter` (`base_plate`), collar ID (`mid_plate`)

- 2 Can height, bracket face to bottom -> base\_plate\_z in outer\_box.scad
- 3 Shaft centre to can centre -> motor\_body\_x\_offset, motor\_x\_offset
- 4 Shaft diameter -> cam bore dia (currently 5.2)
- 5 Shaft flat-to-flat -> cam bore flat clip (currently 3.2)
- 6 Shaft length above the bracket face -> cam hub\_h + shaft\_bore\_depth + z-stack
- 7 Mount-hole centre-to-centre -> motor\_mount\_spacing
- 8 Mount-hole diameter -> motor\_mount\_hole
- 9 Shaft boss (raised collar around the shaft) diameter and height -> cam hub OD (currently dia 9)

## 4. CAM ON SHAFT - clearances OK on paper, z-stack is wrong

### 4a. The bore profile

[SCAD] braille\_cam.scad: intersection(cylinder(r=2.6), cube([10, 3.2, ...])) -> a dia 5.2 cylinder clipped to 3.2 mm across the flats, on both sides (true double-D).

[SPEC] Nominal 28BYJ-48 shaft: **dia 5.0 mm, 3.0 mm across the flats, ~10 mm long.**

- radial clearance:  $(5.2 - 5.0)/2 = 0.10$  mm per side
- flat clearance:  $(3.2 - 3.0)/2 = 0.10$  mm per side

That is correct in intent but *very* tight for a printed part. On resin the bore will come out slightly undersize and will need reaming/filing; on FDM it will be far undersize. Plan on hand-fitting. Do **not** loosen it blindly - a sloppy double-D on a 5.625deg indexing cam translates directly into dot-position error.

### 4b. Engagement depth is 6 mm, not the 8 mm the comment claims

[SCAD] The bore is subtracted over local z = ?5 .. +4 (translate([0, 0, -hub\_h-1]), h = shaft\_bore\_depth + 1 = 9). Material exists only over z = ?4 .. +2 (hub 4 mm + disc floor 2 mm). So:

- **actual maximum engagement = 6.0 mm**, and
- the bore is a **through-hole** - it opens onto the top of the disc floor. There is no ceiling and

therefore **no axial stop** locating the cam on the shaft.

The header comment "Hub 4mm below disc + bore 4mm into disc floor = 8mm total engagement" is stale: the disc floor is 2 mm, not 4 mm.

6 mm of double-D on a ~10 mm shaft is *acceptable* engagement for this torque, but the missing axial stop means cam height depends entirely on how far it is pushed on during assembly - with pin\_lift = 0.8 mm and dot flushness already fought over in v7.2, that is a real risk. Add a 0.5-1.0 mm shoulder at the top of the bore, or a printed depth gauge.

### 4c. The vertical stack does not close - SHOWSTOPPER (conditional on motor measurement)

Working in outer-box world z [SCAD]:

| feature                                              | z               |
|------------------------------------------------------|-----------------|
| motor top / bracket face = base-plate underside      | 41              |
| base plate top                                       | 46              |
| cam pocket floor (base_thickness - cam_pocket_depth) | 43              |
| cam disc floor bottom -> top (cam_flat_z)            | 43 -> <b>45</b> |
| cam hub bottom (4 mm below the disc)                 | <b>39</b>       |
| shaft top, if the shaft is 10 mm [SPEC]              | <b>51</b>       |

Two consequences:

- **The hub is 2 mm too long.** Only 2 mm of space exists between the cam pocket floor (43) and the motor face (41), but hub\_h = 4. The hub bottoms out on the motor and the disc sits **2 mm proud of its pocket**, pushing cam\_flat\_z to 47. Since link\_total\_h is derived from cam\_flat\_z = 45, **every one of the six dots ends up ~2 mm permanently proud** - exactly the class of bug v7.2 just fixed for a 0.8 mm offset.
- **The shaft sticks out the top.** With the bore open, a 10 mm shaft reaches z=51, i.e. **6 mm above the cam surface (4 mm above it even in the mis-seated position). The dot cluster sits at x = +/-2.4, y = 0/+/-2.6 - directly over a dia 5 shaft (r=2.5). The linkage risers occupy 3.5 to 12.2 mm above the cam surface at**

**exactly those coordinates.** The shaft will foul the dot-2 and dot-5 risers.\*\*

**Fixes (all need measurement 6 above first):** shorten hub\_h to the real gap; counterbore the base plate around the shaft so the hub can drop; trim the motor shaft; or lower the motor by increasing the z-budget. Do not guess - measure the shaft, then re-derive the whole stack in one pass.

## 5. ULN2003 IN THE 36 x 46 x 16 POCKET - confirmed

[SCAD] outer\_box.scad: electronics pocket is cube([36, 46, 16]) centred at x=0,y=0, floor at z=4, top at z=20 (X=36, Y=46). mid\_plate.scad adds a through relief slot translate([3, -23, -1]) cube([17, 8, 4]) - 17 mm (x=3..20) by 8 mm (y=-23..-15) - giving **2 mm extra headroom** at that corner only.

| configuration                                           | height [ASSUMED, established earlier] | available                        | verdict                 |
|---------------------------------------------------------|---------------------------------------|----------------------------------|-------------------------|
| vertical Dupont jumpers plugged into the ULN2003 header | ~20 mm                                | 16 mm (18 mm at the relief slot) | <b>FAILS by ~2-4 mm</b> |
| wires soldered flat to the board                        | ~12 mm                                | 16 mm                            | <b>FITS, 4 mm spare</b> |

**Confirmed: solder the wires flat. No vertical jumpers, no exceptions.** This already matches docs/WIRING\_AND\_ASSEMBLY.md.

Three secondary cautions:

**1 Plan fit is tight in X.** A ULN2003 driver board is commonly ~35 x 32 mm [ASSUMED - measure].

If its 35 mm side runs along X it has **0.5 mm per side** in a 36 mm pocket. Orient the 35 mm side along **Y** (46 mm) instead.

**1 Muscle-board bosses are in the way.** Four dia 4 x 4 mm tall bosses stand on the pocket floor at (+/-14, +/-19). If the ULN2003 is ~35 mm in Y (+/-17.5) it clips the y=+/-19 bosses. The pocket is currently spec'd for \*two different boards\* (the future 34x44 muscle PCB and the off-the-shelf ULN2003) and only one of them can be there at a time - decide which.

**1 Floor wire gutters** are cut 2.5 mm deep at x ? [-18,-11] and [11,18] across the whole pocket.

A board laid on the floor will bridge them; fine, but it will not sit flat if it has through-hole tails on the underside.

The mid-plate relief slot is **correctly placed** for a JST/header edge pointed at the (+X, ?Y) corner - it clears the motor collar (whose max x at y=?15 is +0.5) and the boss hole at (26, ?21). Orient the module accordingly.

## 6. ESP32 DEVKIT IN THE POD - FAILS on length and on USB alignment

[SCAD] esp32\_pod\_params.scad / esp32\_pod\_shell.scad: pod\_int\_length = 64 ? 2-4 = 56 -> cavity spans x ? [28, +28]. devkit\_length = 51.5, devkit\_x\_offset = +4.

### 6a. Length - 1.75 mm interference

Board spans x = 4 ? 25.75 = **?21.75** to 4 + 25.75 = **+29.75**. Cavity ends at **+28**. **The board overruns the +X (dock) wall by 1.75 mm.** It will not go in.

The header channels themselves are fine (they span x = ?16 .. +24, retention walls to +/-25.5, all inside the cavity) - it is the \*board overhanging the sockets\* that hits the wall. The antenna grille recess in that wall only starts at z=23, far above the board at z~10.5, so it does not help.

**Fix:** reduce devkit\_x\_offset to ~+2 (giving ~0.25 mm at each end - still too tight) or lengthen the pod. Realistically pod\_length needs to go from 64 to ~68, which conveniently also matches the cell footprint. **Measure the board first.**

### 6b. Width and header pitch - the CAD describes the wrong board

[SCAD] devkit\_width = 28.0, hdr\_row\_pitch = 25.4. The file header says "ESP32 DOIT DevKit V1 (30-pin)".

[SPEC-ish, verify] The commonly cited dimensions are: **30-pin** DOIT V1 ~ 25.4 mm wide with 0.9" (22.86 mm) row pitch; **38-pin** DOIT ~ 28 mm wide with 1.0" (25.4 mm) row pitch. The CAD numbers match the **38-pin** board, while the documentation everywhere in this repo says 30-pin.

If the real board is the 30-pin, the printed header channels are **~2.5 mm too far apart** and the DevKit will not plug in at all. **This is the single most important pod measurement.**

### 6c. USB cutout - ~2 mm too high

[SCAD] `usb_z = pod_floor + hdr_strip_h + 1 = 3 + 8.5 + 1 = 12.5`, cutout 10 wide x 7 tall -> spans  $z = 12.5 \dots 19.5$ .

But `hdr_channel_depth = 1.0` recesses the strips 1 mm into the floor, so the strip top - and therefore the **board underside** - sits at  $3 + 8.5 - 1 = 10.5$ , not 11.5. A micro-USB socket sits on the board and stands ~2.5 mm tall -> it occupies  $z \sim 10.5 \dots 13.0$ .

Overlap with the cutout:  $z = 12.5 \dots 13.0$ , i.e. **0.5 mm**. The port is essentially behind solid wall. `usb_z` should be ~ 10.0.

### 6d. USB cutout size and plug reach

- [SCAD] cutout is 10 x 7 mm in a 4 mm wall.
- [SPEC-ish] a micro-USB plug's \*metal shell\* is ~7 x 2.5 mm, but the **overmoulded body** on a typical cable is ~11-13 mm wide and ~7-9 mm tall. It will very likely not pass a 10 x 7 hole.
- The board's ?X edge sits at  $x = ?21.75$  while the wall inner face is at ?28, so the plug must reach **6.25 mm inboard** past the wall to seat. Combined with the overmould, this is unlikely to work. Enlarge to ~13 x 9 and reduce the standoff distance.

### 6e. Headroom under the lid - fine

Cavity top at  $z = 54$  (`shell_h = 58 - 4`). Board top ~ 10.5, tallest DevKit component maybe 5 mm ->  $z \sim 15.5$ . **~38 mm of unused headroom**. No clash with the jack cradle either: the cradle bottom sits at  $z = 54 - 11 - 1.5 = 41.5$ , and it is at  $y = +18$  while the board spans  $y = +/-14$ .

The pod is enormously over-tall for its contents. Not a fit failure, but worth knowing before printing 58 mm of PETG.

---

## 7. BARREL JACK - UNVERIFIED, and the placeholder cradle already collides

[SCAD] `esp32_pod_lid.scad`:

```
jack_body_w = 9.5;    // PLACEHOLDER
jack_body_l = 12;     // PLACEHOLDER
jack_body_h = 11;     // PLACEHOLDER
cradle_t     = 1.5;
barrel_jack_dia = 11.5; at (x=-22, y=+18)
```

The author has flagged these as placeholders. Two things to know:

**7a. Even the placeholder cradle does not fit the pod.** Cradle outer extent in  $X = barrel\_jack\_x - jack\_body\_l/2 - cradle\_t = ?22 - 6 - 1.5 = ?29.5$ . The pod cavity ends at  $x = ?28$ . **1.5 mm of interference - the lid physically cannot close.** Since a real screw-terminal jack is almost certainly \*bigger\* than the 12 mm placeholder, this gets worse, not better. Either move `barrel_jack_x` inboard (to about ?20) or shrink the cradle.

**7b. The modelled jack is the wrong type.** dia 11.5 is sized for a **threaded panel-mount 5.5/2.1 jack with a nut** [SPEC]. Mridul's part is a **yellow screw-terminal barrel socket** - typically a moulded block with two screw terminals and no panel thread, so a round hole gives it nothing to clamp to. That is exactly why the cradle exists, and exactly why the cradle must be built to the real part.

Also still open from the 2026-06-03 breadboard review: **the jack's polarity has never been proven**. Check it with a multimeter before applying adapter power.

**Measure before touching this file:** see checklist items 14-18 in §10.

---

## 8. MAGNETS - the only genuinely measured component. Re-confirm the new batch.

[SCAD] `outer_box.scad` and `esp32_pod_params.scad` both use `mag_dia = 8.4`, `mag_depth = 1.2`, teardrop profile, 2 per face at  $y = +/-14$ ,  $z = 29$ , in a 4 mm wall (2.8 mm of material left behind each pocket).

[MEASURED, v6.1] `docs/SOURCING.md` records "NeFeB disc magnets **8mm dia x 1mm thick** - plenty" and the v6.1 CAD notes say they were measured off the real parts. So  $8.4 \times 1.2$  gives 0.2 mm radial and 0.2 mm axial glue clearance. **FITS**.

Caveats:

- The bill line "10x magnets + bearing" does **not** identify a size. Confirm the batch in hand is the same 8 x 1 discs that were measured in v6.1, not a different order.
- Quantity: one cell needs **4** (2 per face x +/-X faces); one pod needs **2**. So a cell+pod pair needs 6, and any second cell needs another 4. 10 covers cell + pod + spares, but not a second cell.
- The pockets open onto the **outer** face, so each magnet sits flush with the outside with 2.8 mm of wall behind it. That is the right way round for holding force. A 1 mm-thick 8 mm disc is still a weak magnet - two per face is not much dock retention, so expect the chain to be light-duty.
- The "bearing" on the bill is **no longer needed** - v7.1 replaced the bearing-ball braille dot with a printed 1.5 mm dome (docs/SOURCING.md §3).
- **Separate gap:** the **cam homing magnet (3 x 2 mm)** is a different part and is **not in hand**. See §2c - and note its pocket is currently broken.

## 9. DOT INSERT INTO TOP PLATE - FITS (verified numerically)

[SCAD] From mech\_layout.scad / dot\_insert.scad / top\_plate.scad:

| feature         | insert                              | plate pocket                                                   | clearance    |
|-----------------|-------------------------------------|----------------------------------------------------------------|--------------|
| body (square)   | insert_body = 11.0                  | insert_body + insert_fit = <b>11.2</b>                         | 0.1 mm/side  |
| flange (square) | insert_flange = 15.0                | insert_flange + insert_fit = <b>15.2</b>                       | 0.1 mm/side  |
| body height     | insert_h ?<br>insert_flange_h = 2.0 | opening runs full 4 mm thickness                               | ok           |
| flange height   | insert_flange_h = 1.2               | rebate from z=2.0 to 3.2 = <b>1.2</b>                          | exact        |
| total height    | insert_h = 3.2                      | plate_thickness ?<br>finger_pad_depth = 4.0 ? 0.8 = <b>3.2</b> | <b>flush</b> |

- **Glue shelf:** rebate floor at z = 2.0, ring width  $(15.2 - 11.2)/2 = 2.0$  mm all round, area =  $15.22 - 11.22 = 105.6$  mm<sup>2</sup>. Matches the documented ~106 mm<sup>2</sup>.
- **Spring recess ceiling:** spring\_bores cut z = ?0.01 .. 2.0, i.e. exactly the body height, so the ceiling is the flange underside at insert-z 2.0 = linkage-y 11.0. This matches the mech\_layout.scad comment "dot DOWN 8.0 -> 11.0 = 3.0 mm". **Consistent.**
- **Dot clearance:** plate\_hole\_dia 1.7 vs dot\_dome\_dia 1.5 -> 0.1 mm/side. Correct.
- **Feature envelope:** dots span x = +/-2.4, y = +/-2.6; plus the dia 2.2 spring bore radius that is x = +/-3.5, y = +/-3.7 - comfortably inside the 11 mm body.

**Verdict: FITS.** The only real risk is manufacturing, not geometry:

- 1 The plate is PETG on a 0.4 mm nozzle. A square 11.2 mm hole prints undersize and picks up elephant's foot on the first layers. Expect to **file the opening** or add a 0.15-0.2 mm chamfer.
- 1 0.1 mm/side is a \*press\* fit, not a \*glue\* fit. Thin CA will wick in; epoxy will not - it needs ~0.15-0.2 mm. Use thin CA on the shelf, per the assembly note in dot\_insert.scad.
- 1 Wipe squeeze-out off the top face before it cures or the domes jam. Already documented.

## 10. MEASUREMENTS REQUIRED BEFORE ASSEMBLY

Every one of these must be taken with calipers on **the actual part in hand**, not from a datasheet or a product listing. The right-hand column is the CAD symbol it feeds.

**Motor - 28BYJ-48 (blocks base\_plate, mid\_plate, braille\_cam, outer\_box)**

| # | Measure            | Feeds                                                        |
|---|--------------------|--------------------------------------------------------------|
| 1 | Can outer diameter | motor_body_diameter (base_plate); collar ID 29.5 (mid_plate) |

| #  | Measure                                        | Feeds                                                  |
|----|------------------------------------------------|--------------------------------------------------------|
| 2  | Can height, bottom face to bracket face        | base_plate_z = 41 (outer_box)                          |
| 3  | Shaft centre -> can centre distance            | motor_body_x_offset = ?8;<br>motor_x_offset = ?8       |
| 4  | Shaft diameter                                 | cam bore cylinder dia 5.2 (r=2.6, braille_cam)         |
| 5  | Shaft flat-to-flat width                       | cam bore clip cube( [10, 3.2, ...] )                   |
| 6  | Shaft length above the bracket face            | hub_h = 4, shaft_bore_depth = 8, and the whole z-stack |
| 7  | Shaft boss (raised collar) diameter and height | cam hub OD dia 9 (r=4.5)                               |
| 8  | Mount-hole centre-to-centre                    | motor_mount_spacing = 35                               |
| 9  | Mount-hole diameter                            | motor_mount_hole = 4.3                                 |
| 10 | Bracket ear outer width / thickness            | base_length = 56 (the 0.35 mm edge wall problem)       |

### *Hall sensor (blocks base\_plate)*

| #  | Measure                                                                | Feeds                                                                         |
|----|------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 11 | MH-Sensor module PCB length x width x thickness                        | hall_pocket_w = 5.3, hall_pocket_d = 4.3 - <b>or</b> the decision to desolder |
| 12 | Header height above the PCB (or below, if it must lie flat)            | hall_pocket_h = 3                                                             |
| 13 | If desoldering: bare TO-92 body width x depth x height, and lead pitch | hall_pocket_w/d/h, hall_wire_w/d                                              |

### *Barrel jack (blocks esp32\_pod\_lid)*

| #  | Measure                                                                | Feeds                                       |
|----|------------------------------------------------------------------------|---------------------------------------------|
| 14 | Jack body X length (along the pod's X axis)                            | jack_body_l = 12 (PLACEHOLDER)              |
| 15 | Jack body Y width                                                      | jack_body_w = 9.5 (PLACEHOLDER)             |
| 16 | Jack body Z height below the lid, including screw terminals            | jack_body_h = 11 (PLACEHOLDER)              |
| 17 | Barrel outer diameter at the face (the part that goes through the lid) | barrel_jack_dia = 11.5                      |
| 18 | Does it have a panel thread + nut? Thread dia and nut AF if so         | decides hole vs cradle strategy             |
| 19 | <b>Polarity</b> (multimeter: which terminal is centre-pin positive)    | wiring, not CAD - but do it before power-on |

### *ESP32 DevKit (blocks esp32\_pod\_shell, esp32\_pod\_params)*

| #  | Measure                                                                | Feeds                                |
|----|------------------------------------------------------------------------|--------------------------------------|
| 20 | Board length                                                           | devkit_length = 51.5                 |
| 21 | Board width                                                            | devkit_width = 28.0                  |
| 22 | Pin-row centre-to-centre pitch ( <b>critical - 30-pin vs 38-pin</b> )  | hdr_row_pitch = 25.4                 |
| 23 | Pin count per row                                                      | hdr_strip_len = 40                   |
| 24 | USB socket: width, height, and height of its underside above the board | usb_w = 10, usb_h = 7, usb_z = 12.5  |
| 25 | Your USB cable's overmould width x height                              | usb_w, usb_h                         |
| 26 | Female header strip: body width and height with a board seated         | hdr_strip_w = 2.7, hdr_strip_h = 8.5 |

### *ULN2003 driver module (blocks outer\_box, mid\_plate)*

| #  | Measure                                           | Feeds                                           |
|----|---------------------------------------------------|-------------------------------------------------|
| 27 | Board length x width                              | elec_pocket 36 x 46 (outer_box)                 |
| 28 | Total height <b>with wires soldered flat</b>      | elec_pocket_h = 16                              |
| 29 | Height and position of the tallest connector      | mid_plate relief slot [3, ?23]<br>cube([17, 8]) |
| 30 | Mounting-hole positions, if you intend to bolt it | mb_boss_positions = (+/-14, +/-19)              |

### *Magnets and switches*

| #  | Measure                                                                   | Feeds                                                              |
|----|---------------------------------------------------------------------------|--------------------------------------------------------------------|
| 31 | Docking magnet diameter x thickness (re-confirm the new batch)            | mag_dia = 8.4, mag_depth = 1.2                                     |
| 32 | Homing magnet diameter x thickness ( <b>must be bought - 3x2 mm</b> )     | magnet_dia = 3.0, magnet_depth = 2.0                               |
| 33 | Tactile switch body W x D, total height including plunger, plunger travel | sw_body = 6.4, sw_depth = 5.5, cage depth 7.2, nav_shaft_len = 4.5 |
| 34 | Switch lead spread and lead orientation                                   | switch cage fin spacing                                            |

### *Pogo pins (not audited here - no dimensions exist in the CAD at all)*

| #  | Measure                                       | Feeds                                                                                                     |
|----|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 35 | Pogo carrier board width x height x thickness | pogo_carrier_w = 12, pogo_carrier_h = 10, pogo_carrier_d = 2 - all explicitly marked "measure real part!" |
| 36 | Pogo pin free length and compressed length    | pogo_slot_cube([6, 10, 8]),<br>pogo_pad_recess = 1                                                        |

## 11. BOTTOM LINE

**Do not order resin yet, and do not attempt a full assembly.**

Blocking failures that are certain from the source, independent of any measurement:

- 1 The hall sensor pocket **does not exist** - it is entirely inside the cam pocket at the same depth (§2a).
- 2 The cam homing-magnet pocket is a **through-hole that craters cam tracks 2 and 3** (§2c).
- 3 The motor's **right mounting screw hole opens into the spring cavity** - one-screw motor (§3a).
- 4 The motor's **left screw hole has a 0.35 mm wall** to the plate edge (§3b).
- 5 The **ESP32 board overruns the pod cavity by 1.75 mm** (§6a).
- 6 The **USB cutout is ~2 mm above the port** (§6c).
- 7 The **jack cradle overruns the pod cavity by 1.5 mm**, with placeholder dimensions (§7a).
- 8 The **cam z-stack does not close**: hub 2 mm too long, shaft ~6 mm too long for the open bore,

putting all six dots permanently proud and the shaft into the dot-cluster risers (§4c).

What actually passes: the **dot insert / top plate** interface (§9), the **docking magnets** (§8), and the **ULN2003 in the pocket provided its wires are soldered flat** (§5).

The single highest-value next action remains the one in the handoff: **take the 36 measurements above**, starting with the motor, then re-derive the z-stack in one pass. Nothing downstream is trustworthy until the motor is a measured object rather than a datasheet.